

Multiple Choice

1. **Answer:** A

Options: A) 400 Mpa; B) 600 Mpa; C) 750 Mpa; D) 150 Mpa; E) 500 Mpa

Note: Correct option: A - 400 Mpa

2. **Answer:** C

Options: A) $y(x) = c_1 * \sin(x) + c_2 * \cos(x)$; B) $y(x) = c_1 * y_1(x)^2 + c_2 * y_2(x)^2$; C) $y(x) = c_1 y_1(x) + c_2 y_2(x)$; D) $y(x) = c_1 * x * y_1(x) + c_2 / (x * y_2(x))$; E) $y(x) = c_1 * y_1(x) / (c_2 * y_2(x))$

Note: Correct option: C - $y(x) = c_1 y_1(x) + c_2 y_2(x)$

3. **Answer:** B

Options: A) $(S^2 / 3)$; B) $(S^2 / 12)$; C) $(S^2 / 24)$; D) $(S^2 / 8)$; E) $(S^2 / 18)$

Note: Correct option: B - $(S^2 / 12)$

4. **Answer:** E

Options: A) $L\{t^n u(t)\} = [n! / (s^{(n+3)})]$; B) $L\{t^n u(t)\} = [(n+1)! / (s^{(n+1)})]$; C) $L\{t^n u(t)\} = [n! / (s^n)]$; D) $L\{t^n u(t)\} = [n! / (s^{n+2})]$; E) $L\{t^n u(t)\} = [n! / (s^{n+1})]$

Note: Correct option: E - $L\{t^n u(t)\} = [n! / (s^{n+1})]$

5. **Answer:** A

Options: A) 0.30; B) 0.92; C) 0.99; D) 1.05; E) 0.65

Note: Correct option: A - 0.30

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'reactance.'

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'zero.'

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $f(t) = 1 + 2 \sum_{k=-\infty}^{\infty} k = 1[\{ \cos kt + k \sin kt \} / \{ 1+k^2 \}]$ '.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '27'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'resolution.'

Long Form Response

11. **Answer:** 3.12 in. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 3.12 kN. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key